BAJAJ ALLOYS, BHIWANI

VALVES

Complete Industry & Product Report: Product, Types, Manufacturing, Market, B2B / B2C

Prepared by: Intern, Business Research
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1. Executive summary

A valve is a mechanical device that starts, stops, regulates or directs the flow of liquids, gases, steam or slurries through a pipe. Valves are used in almost every industry: water supply, oil & gas, power, chemicals, pharma, food, HVAC, plumbing and the home. For a brass and copper-alloy foundry such as Bajaj Alloys, valves are one of the largest end-products for cast and forged brass: bodies, bonnets, stems, ball and seat parts are all made from brass, bronze and gunmetal.

- **Market:** India industrial valves was about USD 3.7 billion in 2024 and is forecast at about USD 5.9 billion by 2030 (about 7.7% CAGR, TechSci Research). Other publishers give USD 2 to 5.8 billion because of different scope.
- **Where brass fits:** Small and medium valves (up to about 4 inch / DN100) for water, plumbing, gas, HVAC and fire-fighting are mostly brass or bronze. Large process valves are mostly cast iron, steel or stainless steel.
- **Channels:** B2B (OEMs, contractors, industry, government) is the larger part of value. B2C (plumbing retail, hardware shops, online) is smaller in value but grows steadily with housing.
- **Recommendation:** Start with brass ball valves, gate valves and cocks for plumbing and agricultural use, and sell through distributors (B2B2C); then move to certified fire and gas valves and OEM supply.

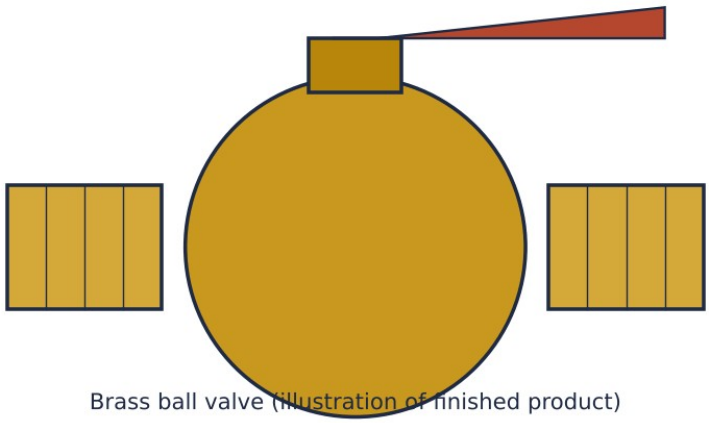


Figure 1: Brass ball valve, illustration of a typical finished product

2. What is a valve?

A valve is a flow-control device. Inside it, a movable element (gate, disc, ball, plug, flap or diaphragm) opens, closes or partly blocks a passage. An operator (hand lever, wheel, gear, motor, air cylinder or solenoid) moves that element.

Main parts of a valve

Table 1: Valve parts and materials

Part	Function	Usual material (brass valve)
Body	Pressure-holding shell with the flow passage	Forged or cast brass (CW617N, IS 292 / IS 318 grades)
Bonnet / cover	Closes the body and holds the stem	Brass

Stem / spindle	Transfers motion from handle to closure element	Brass, dezincification-resistant brass, stainless steel
Closure element	Gate, ball, disc or plug that stops the flow	Chrome-plated brass ball, bronze disc
Seat / seal	Gives leak-tight sealing	PTFE, nylon, rubber, metal seat
Handle / actuator	Moves the stem	Zinc/Al alloy, steel, or motorised/pneumatic
Gland / packing	Stops leakage around the stem	PTFE, graphite, O-ring

Why valves matter

- Isolation: shut off a section for repair.
- Regulation: control flow rate or pressure.
- Protection: stop back-flow or release over-pressure.
- Distribution: divert flow in different directions.

3. Types of valves

Valves are classified by motion of the closure element and by function.

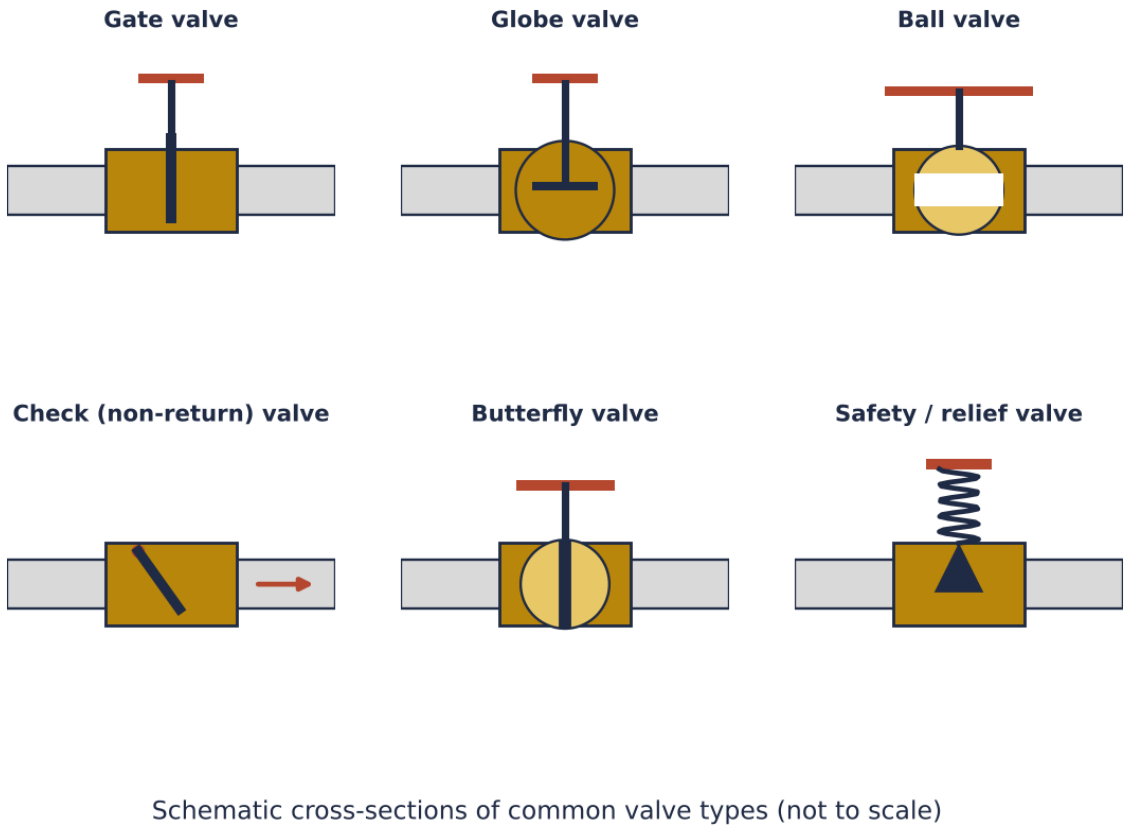


Figure 2: Cross-section schematics of six major valve types

Table 2: Comparison of common valve types

Valve type	Motion	Best use	Typical material	Size range
Gate valve	Linear (lift)	On/off isolation, low pressure drop	Brass, bronze, CI, steel	15 mm to 600 mm+

Globe valve	Linear	Throttling and flow control	Brass, bronze, steel	15 mm to 300 mm
Ball valve	Quarter-turn	Quick on/off, gas, water, LPG	Forged brass, SS, CS	6 mm to 600 mm
Check valve	Self-acting	Prevents back-flow (pumps)	Brass, CI, SS	15 mm to 600 mm
Butterfly valve	Quarter-turn	Large water lines, HVAC	CI, ductile iron, SS	50 mm to 3000 mm
Plug / cock	Quarter-turn	Simple taps, gas cocks	Brass	8 mm to 100 mm
Safety / relief valve	Self-acting (spring)	Over-pressure protection (boilers, compressors)	Bronze, steel	15 mm to 300 mm
Solenoid valve	Electrical	Automation, appliances	Brass body + coil	6 mm to 50 mm
Needle valve	Linear fine thread	Precise small-flow control, instrumentation	Brass, SS	6 mm to 25 mm
Diaphragm valve	Linear	Pharma, food, corrosive media	SS, lined	15 mm to 200 mm

By application

- **Plumbing & sanitary:** bib cocks, angle cocks, stop cocks, ball valves, float valves
- **Agriculture & irrigation:** foot valves, sluice valves, air valves, ball valves
- **Industrial process:** gate, globe, control and safety valves
- **Fire-fighting:** landing valves, hydrant valves, alarm check valves
- **Gas & LPG:** gas cocks, regulators, safety shut-off valves
- **Automotive & appliances:** solenoid valves, engine valves, water-heater valves

By material

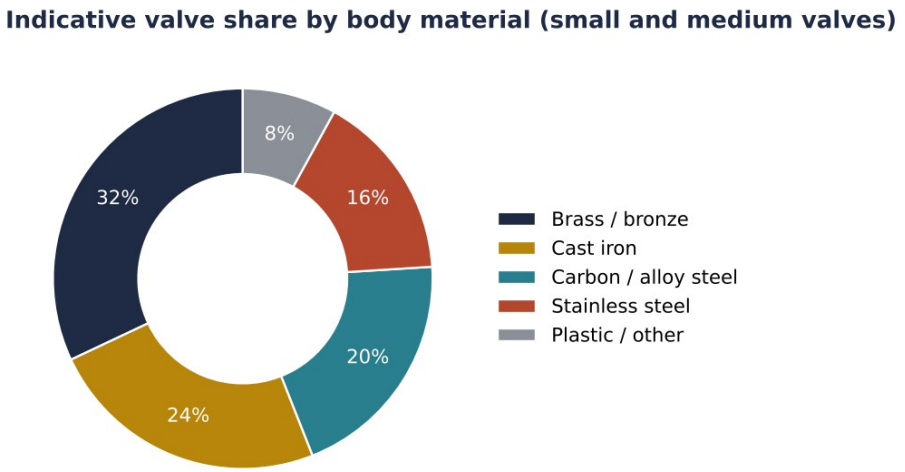


Figure 3: Indicative split of body materials (analyst estimate, intern compilation)

4. Materials: where brass and copper alloys fit

Table 3: Common copper alloys for valves

Alloy	Typical composition	Why used	Used for
Free-cutting brass (CW614N / IS 319)	Cu 58 to 60, Zn 38 to 40, Pb 2 to 3	Easy machining	Cocks, fittings, stems
Forging brass (CW617N)	Cu 57 to 59, Zn rest, Pb about 2	Hot-forges well	Ball valve bodies
DZR brass (CW602N)	Cu 61 to 63, As added	Resists dezincification	Potable water, hot water
Leaded gunmetal (IS 318 LTB2)	Cu 85, Sn 5, Pb 5, Zn 5	Pressure tight casting	Gate/globe valve bodies
Aluminium bronze	Cu 80 to 88, Al 8 to 11	Strength, sea water resistance	Marine valves

Because brass is recycled readily, Bajaj Alloys' existing scrap procurement and melting set-up directly feeds ingots and rods for this product line.

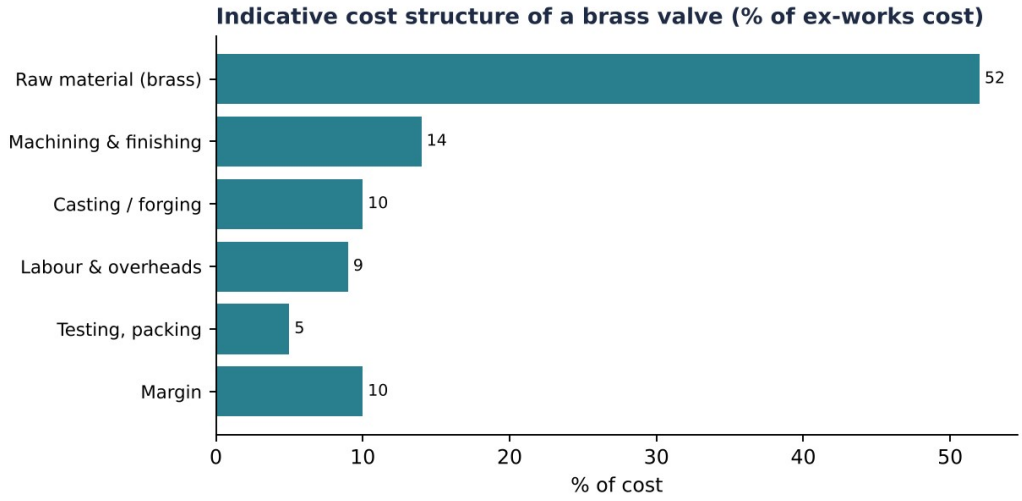


Figure 4: Indicative cost structure (illustrative, to be validated with plant data)

5. Manufacturing process

Brass valves are made either by sand/gravity die casting (gunmetal, gate and globe valves) or by hot forging (ball valve bodies, cocks).

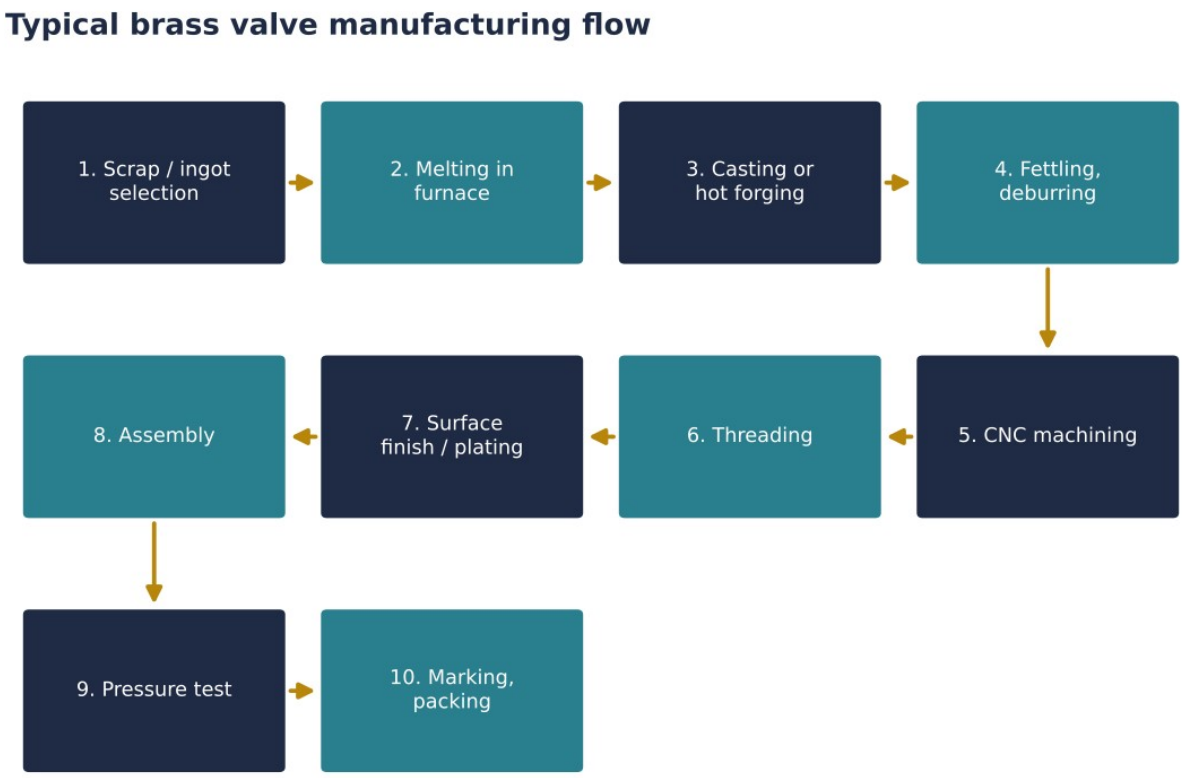


Figure 5: Process flow for a brass valve

- **Step 1: Raw material and melting:** Cu and Zn (or brass scrap) are charged in an induction or crucible furnace at 950 to 1050 degrees C. Composition is checked by spectrometer, then degassed and poured.
- **Step 2: Casting or forging:** Casting: molten metal into sand or permanent moulds, then cooled. Forging: brass billets are heated to about 750 degrees C and pressed in dies for denser, leak-free bodies.
- **Step 3: Fettling and heat treatment:** Runners/risers are cut, flash is removed by shot blasting; stress-relief annealing for forged parts.
- **Step 4: Machining:** CNC/VMC and transfer machines bore the body, cut threads (BSP/NPT), face seats and turn stems and balls.
- **Step 5: Surface treatment:** Pickling, polishing, nickel/chrome plating or lacquer.
- **Step 6: Assembly:** Seat, ball/disc, stem, O-rings, gland nut and handle are fitted on assembly lines.
- **Step 7: Testing:** 100% hydrostatic shell and seat test, and air-under-water test for leak-tightness.
- **Step 8: Marking, packing and dispatch:** Size, pressure rating (PN), brand and batch marking, then boxing.

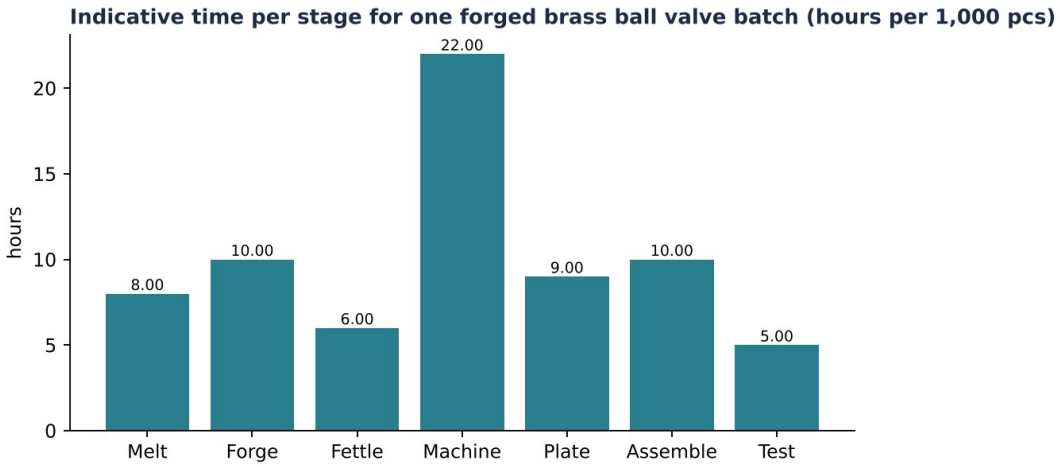


Figure 6: Indicative production time per 1,000 pieces (illustrative)

6. Quality testing and standards

Table 4: Standards relevant to brass valves

Standard / test	Scope
IS 778	Copper alloy gate, globe and check valves for water works
IS 1703	Copper alloy float valves
IS 319	Free-cutting brass bars and sections
IS 318	Leaded tin bronze ingots and castings
BIS (ISI) mark	Mandatory certification for many plumbing valves
API 598 / EN 12266	Valve pressure testing
ISO 9001	Quality management system
Lead-free / DZR requirement	Needed for potable water exports (EU, US)

Typical in-house tests: spectro chemical analysis, hardness, dezincification test, hydrostatic test, torque test, salt-spray for plating, and dimensional check.

7. Market size and growth

7.1 India

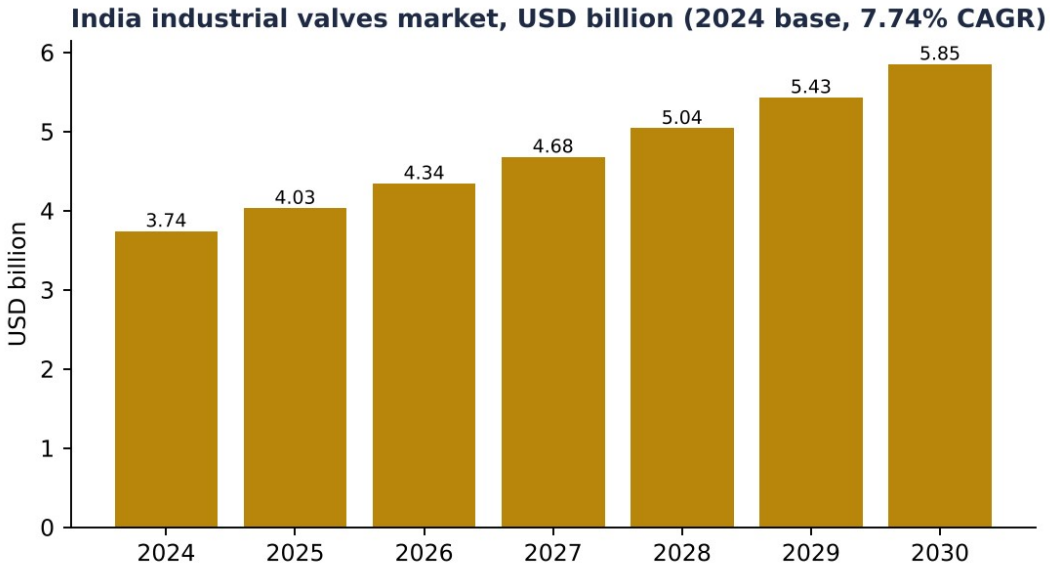


Figure 7: India industrial valves market forecast

Table 5: Published estimates differ because of scope

Publisher	Base year value	CAGR / forecast	Scope note
TechSci Research	USD 3.74 bn (2024)	7.74% to USD 5.90 bn (2030)	Industrial valves
Verified Market Research	USD 3.74 bn (2024)	USD 6.1 bn by 2032	Industrial valves
Data Bridge	USD 5.84 bn (2024)	About 8.96% to USD 9.97 bn (2032)	Industrial valves, wider scope
IMARC (industrial)	USD 3.4 bn (2025)	6.64% (2026 to 2034)	Industrial valves
IMARC (valves)	USD 2.0 bn (2024)	3.5% (2025 to 2033)	Valves, narrower definition
Mordor Intelligence	USD 1.56 bn (2025)	6.96% (2026 to 2031)	Narrower segments

Take-away: all publishers agree on steady mid-to-high single-digit growth. A planning range of USD 3 to 4 billion today and 6 to 7% growth is reasonable.

7.2 Demand drivers

- Government water programmes (Jal Jeevan Mission, AMRUT) need millions of taps, valves and meters.
- Housing, real estate and Smart-City construction.
- Oil & gas, city-gas distribution and power capacity addition.
- Irrigation and agri-pumping.
- Chemicals, pharma and food capacity expansion; PLI schemes and 'Make in India'.
- Replacement demand from ageing plants.

7.3 Segmentation (indicative)

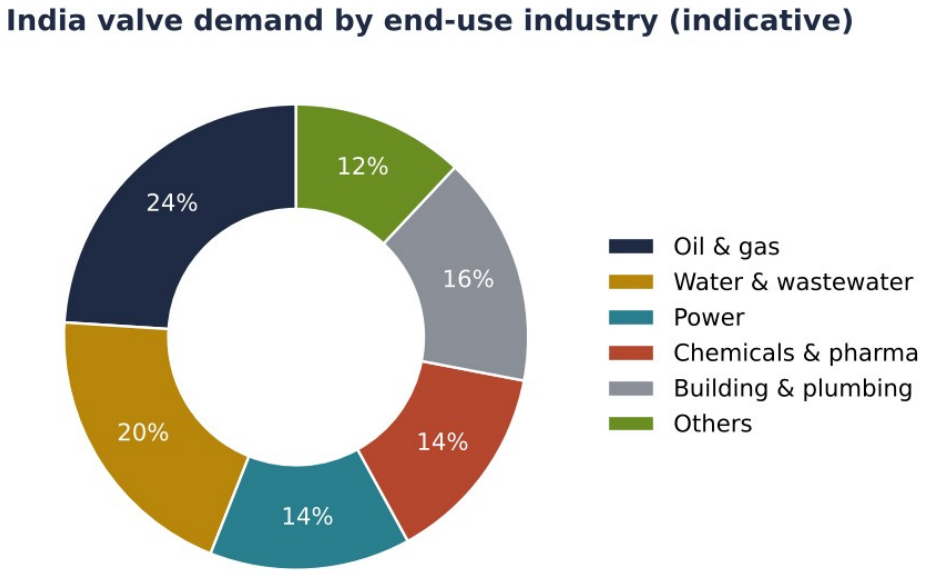


Figure 8: Indicative end-use split (analyst-style estimate; to be validated)

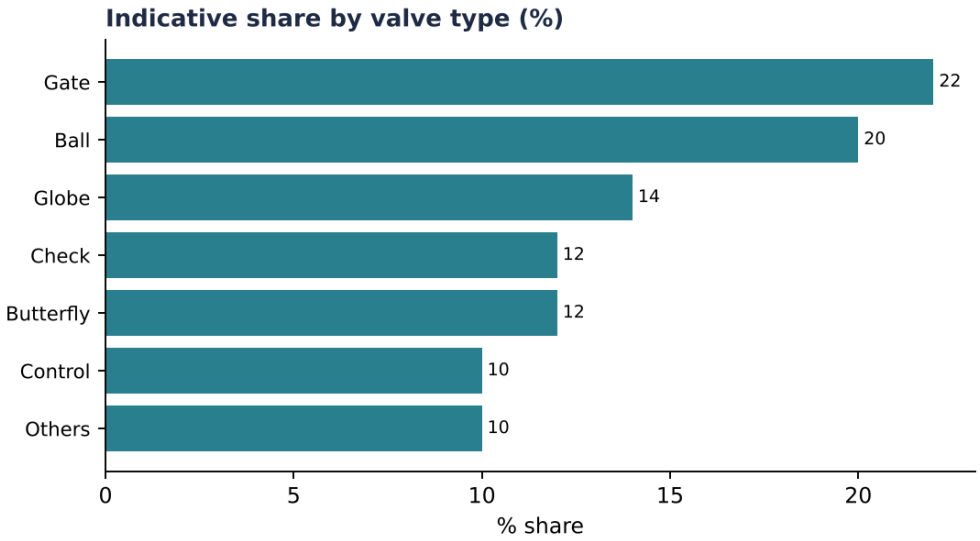


Figure 9: Indicative share by valve type

7.4 Regional clusters in India

- **Jamnagar (Gujarat):** brass parts and forgings hub
- **Coimbatore (Tamil Nadu):** pumps and industrial valves
- **Pune and Mumbai:** process valves, OEM
- **Jagadhri (Haryana):** brass and utensil / valve cluster near Bhiwani
- **Delhi NCR, Faridabad, Bhiwani:** brass casting, scrap recycling, fittings

7.5 Global picture

The global valve market is large (published figures are in the tens of billions of USD, roughly USD 80 to 100 billion depending on publisher and scope) and grows at about 4 to 6% a year. Asia-Pacific is the biggest and fastest growing region. India is a rising source for brass valves, forged parts and castings for Europe and the USA. These global numbers were not verified in this research round and should be confirmed before use.

8. B2B and B2C market

8.1 Definitions

- **B2B (business to business):** Valves sold to OEMs, EPC contractors, industries, government departments and distributors/wholesalers.
- **B2C (business to consumer):** Valves sold to households and small plumbers through hardware and sanitary shops, e-commerce and brand showrooms.

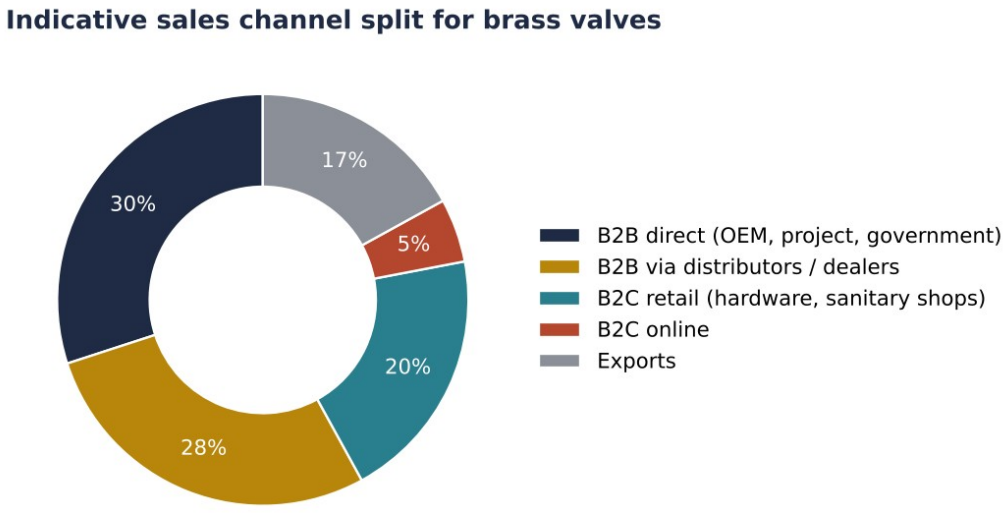


Figure 10: Indicative channel mix for a brass valve maker (illustrative)

Table 6: B2B vs B2C comparison for valves

Factor	B2B	B2C
Buyers	OEMs, contractors, industry, PSUs, distributors	Households, plumbers, small shops
Order size	Large, repeat, tender-based	Single pieces, small lots
Price sensitivity	High, tender L1 rates	Medium; brand and look matter
Quality need	Standards, test certificates, BIS/API	Finish, brand, warranty
Payment	Credit 30 to 90 days, LC	Cash/UPI quickly via retailer
Margin	Lower (8 to 15%)	Higher (20 to 35% brand margin)
Marketing	Key-account sales, tender, samples	Dealer network, mechanic schemes, digital marketing
Examples	L&T, Kirloskar, Larsen, BHEL supply chains; state water boards	Hardware shops, Amazon, IndiaMART retail

8.2 Digital B2B platforms

IndiaMART, TradeIndia, Alibaba and Udaan are active lead-generation channels for brass valves from Jamnagar and Haryana manufacturers.

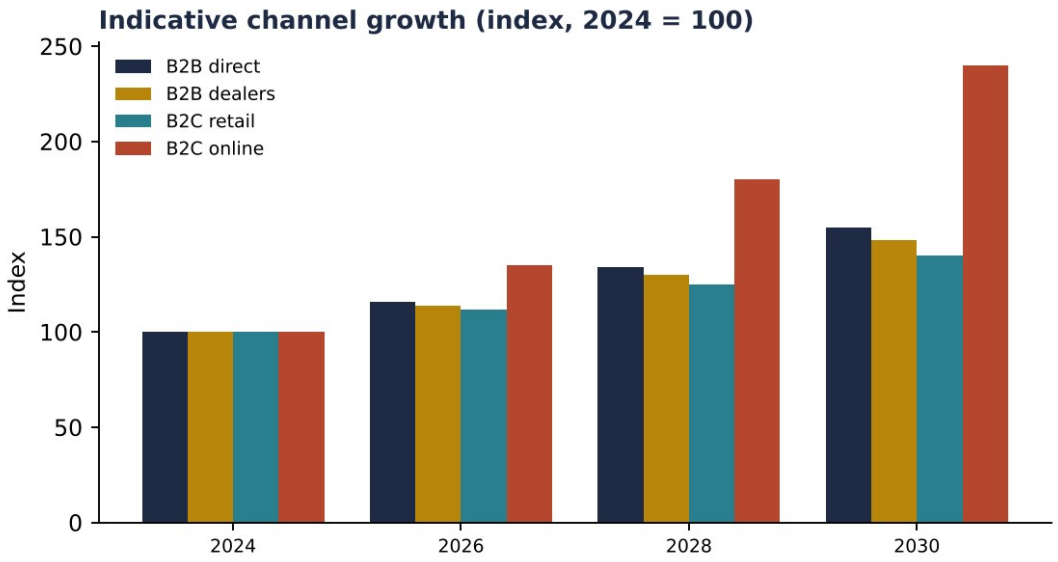


Figure 11: Illustrative growth: online grows fastest from a small base

9. Competition

Table 7: Competitive landscape (names for orientation only)

Segment	Players (examples)
Organised industrial valves	L&T Valves, Kirloskar Brothers, Audco (L&T), BEL Valves, Advance Valves, Fouress, Leader Valves
Brass / plumbing valves	Zoloto, Leader, Sant, Kitz-type imports, Jamnagar and Jagadhri small units
Global	Emerson, Flowserve, Crane, KITZ, Cameron/SLB, Spirax-Sarco

The brass valve segment is fragmented: hundreds of small units. Competition is on price, but certified quality and consistent supply are rare, which is the gap for a quality-focused foundry.

10. Opportunity and recommendations for Bajaj Alloys

- Phase 1: forged brass ball valves and cocks 1/2 to 2 inch. Use own brass rods and billets, forge, CNC machine and pressure test.
- Phase 2: gunmetal gate/globe valves and foot valves for irrigation and water boards (IS 778).
- Phase 3: DZR, lead-free grades for export and potable water; fire and gas valves with certification.
- Sell to distributors first (B2B2C), build a brand in Haryana/Rajasthan/Punjab, add IndiaMART and Amazon Business.
- Get BIS, ISO 9001 and a hydraulic test bench before approaching tenders.
- Risks: copper/zinc price swings, imports from China, working-capital in credit sales, lead regulations.

Table 8: Indicative roadmap

Item	Action	Timeline
Product range finalisation	Choose 5 SKUs, costing	Month 1
Tooling and dies	Forging dies, jigs	Months 2 to 4
Certification	BIS, ISO	Months 3 to 8
Pilot batch and test	Customer trials	Month 5
Dealer network	20 distributors	Months 6 to 12

11. Data notes and sources

Market-size values for India come from published research summaries (TechSci Research, Verified Market Research, Data Bridge Market Research, IMARC Group, Mordor Intelligence). Figures differ because of scope; intermediate yearly values in charts are interpolated at the stated CAGR. Charts labelled 'indicative' (material share, end-use, valve type, channel mix, cost structure, production time) are intern estimates built from typical industry structure; they are NOT audited statistics and must be validated with plant and customer data before management decisions. Diagrams are schematic drawings made for this report, not photographs; replace them with company photographs where needed.

- TechSci Research: India Industrial Valves Market
- IMARC Group: India Valves Market, India Industrial Valves Market
- Data Bridge Market Research: India Industrial Valves Market to 2032
- Verified Market Research: India Industrial Valve Market
- Mordor Intelligence: India valves market
- Bureau of Indian Standards: IS 778, IS 318, IS 319